## Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Execute the Coursera Lab Jupyter notebook for this lesson, which implements the Monte-Carlo method to approximate an integral. This is the same example presented in the lesson.

   1 / 1 point

   What is the root-mean-squared error between the true integral value and the estimate produced by the Monte-Carlo method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

   Note: Make sure that the code is configured with the default values of `N=10000` and `sx = 0.75` if you have changed them to experiment. Also make sure that the code executes `randn("state",0)` before generating random numbers, since otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

   0.0049

   **Correct**
   Yes. That is correct.

2. For the same Coursera Lab Jupyter notebook, change the value of `sx` from the default of `0.75` to the new value of `1.0`. Do not modify anything else from the default conditions.

   1 / 1 point

   What is the root-mean-squared error between the true integral value and the estimate produced by the Monte-Carlo method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

   0.0063

   **Correct**
   Yes, that is correct.

3. For the same Coursera Lab Jupyter notebook, change the value of `N` from the default of `10000` to the new value of `25000`. Do not modify anything else from the default conditions. (Make sure that you have changed `sx` back to the default value of `0.75`, for example.)

   1 / 1 point

   What is the root-mean-squared error between the true integral value and the estimate produced by the Monte-Carlo method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

   0.0017

   **Correct**
   Yes, that is correct.